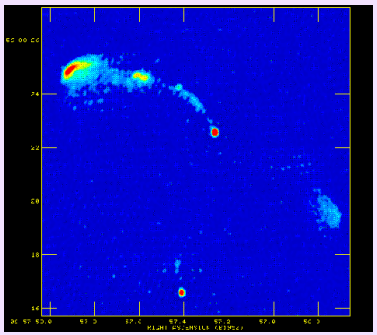
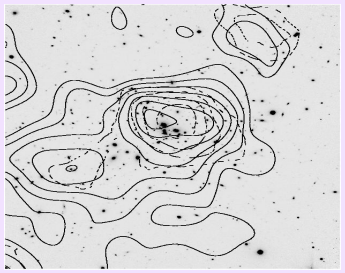






(STRONG!)

Gravitational Lensing



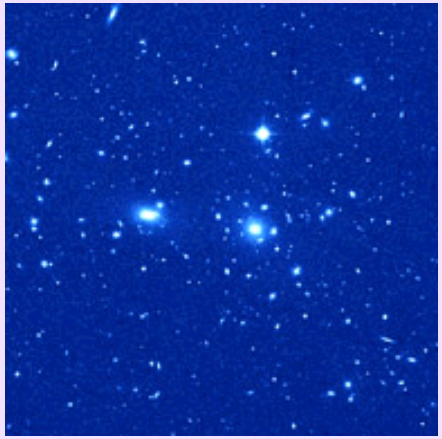




Recall “Classical” evidence for dark matter



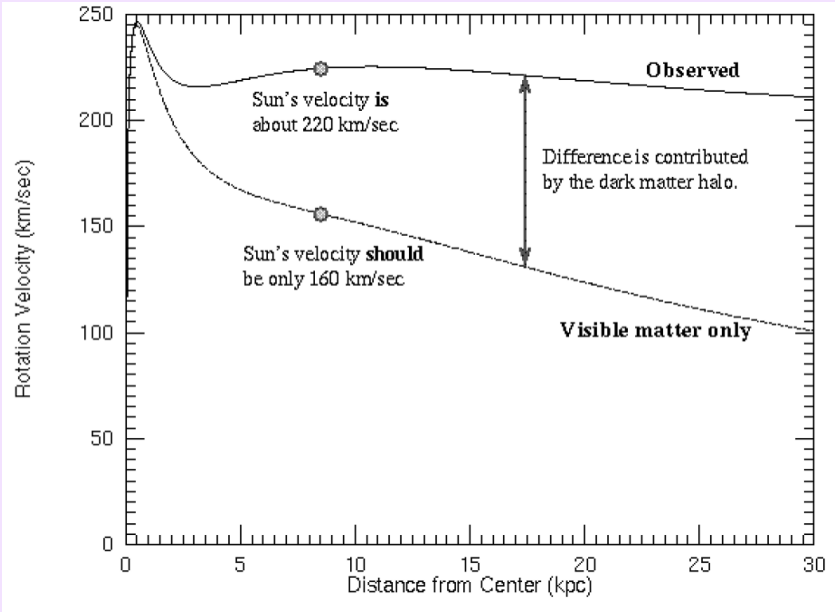
Fritz Zwicky



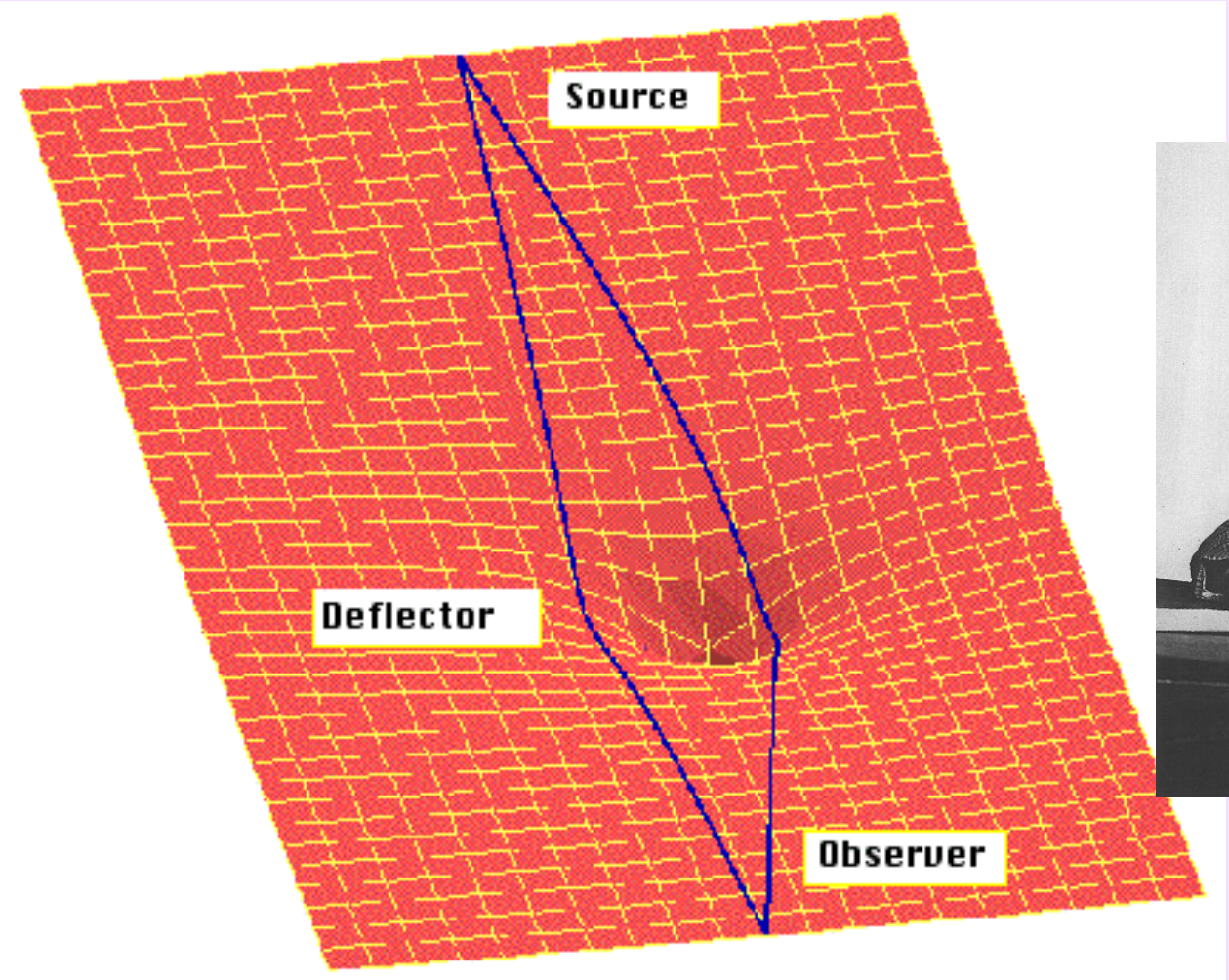
Coma Cluster

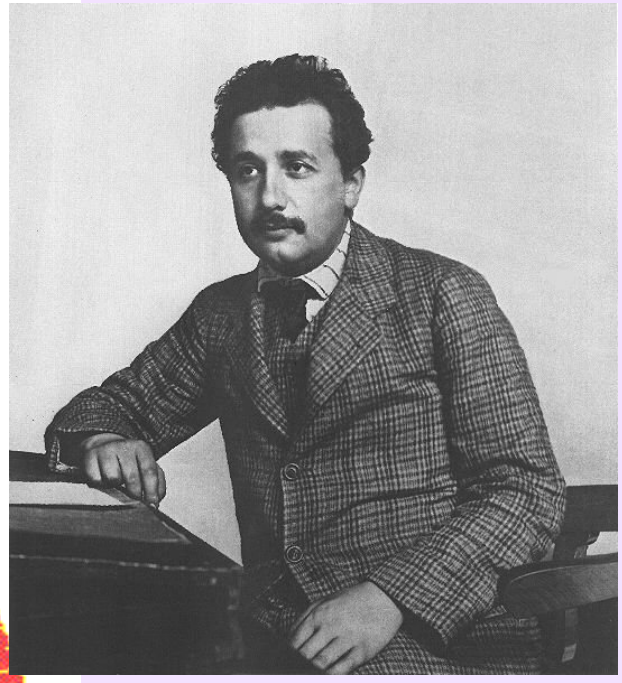
Dynamics of cluster galaxies - ~100 times more mass than implied by luminous matter. Accounting for x-ray gas - still ~10 times more mass required

Galaxy rotation curves

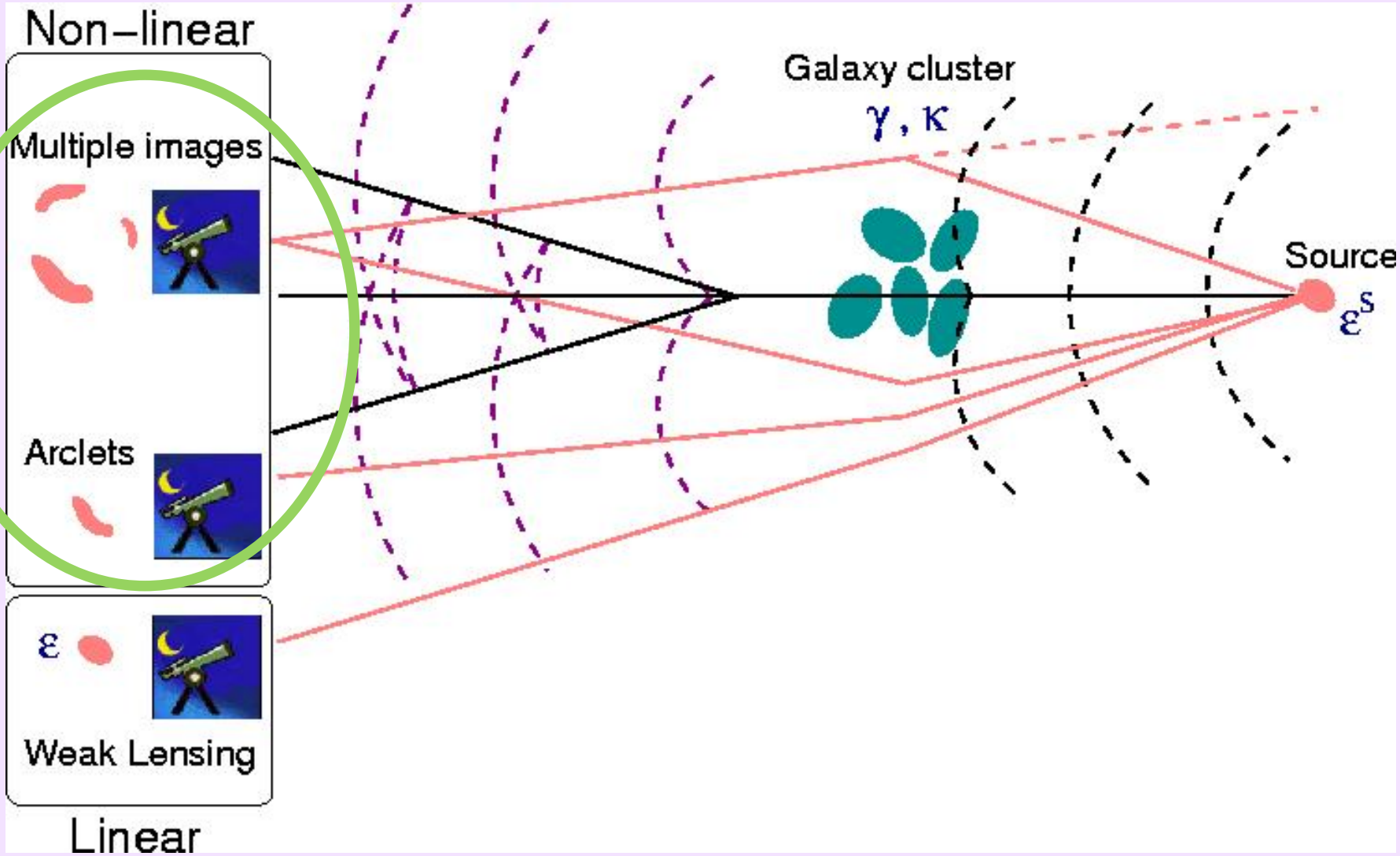


Einstein's General Relativity: mass distorts space-time - light is deflected - "gravitational lensing"

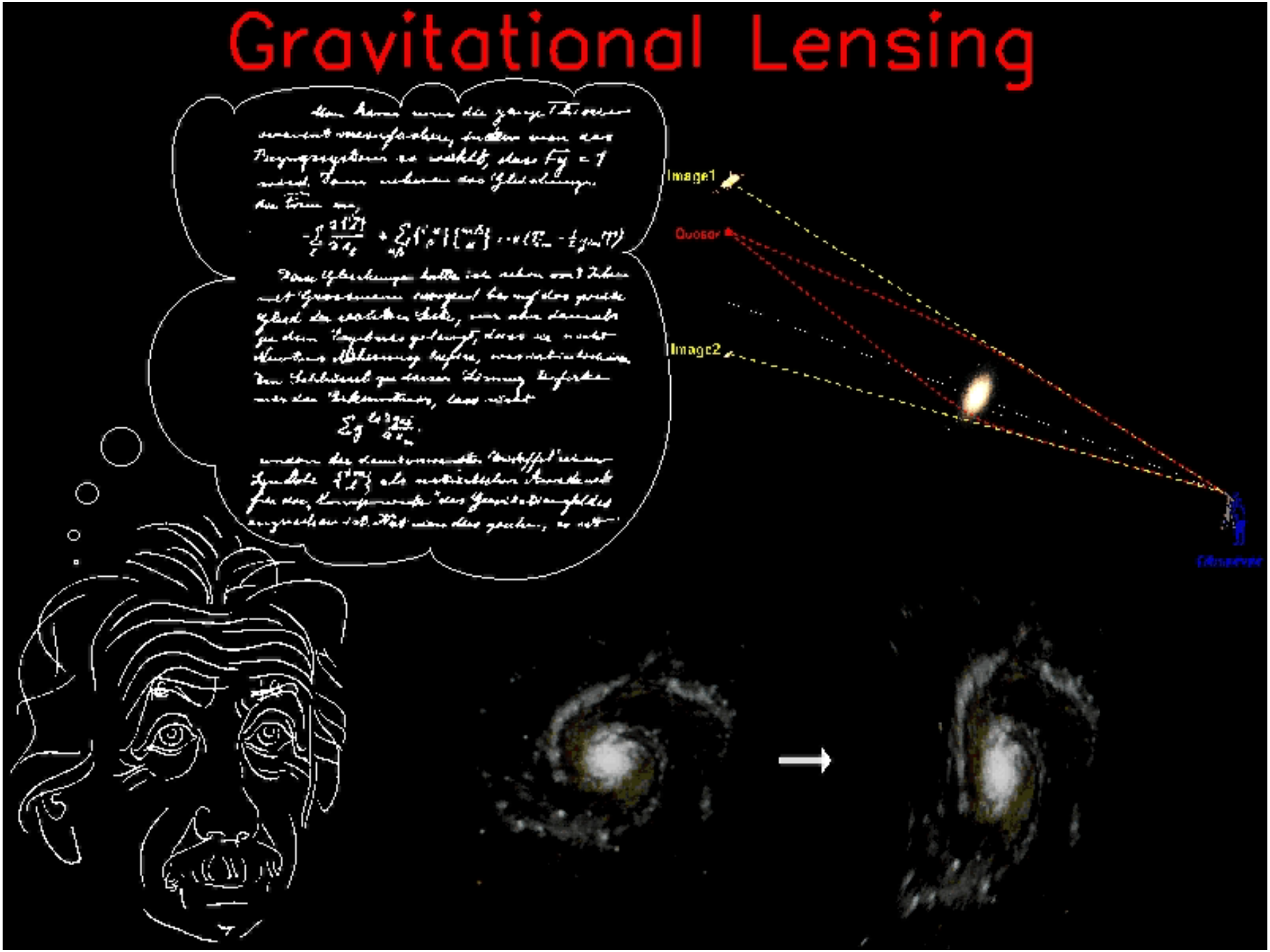




Regimes of gravitational lensing



Gravitational Lensing



Some historical milestones

- 1804 Soldner: first written account of deflection of light by gravity - light ray close to solar limb deflected by 0.84 arcsec

- 1911 Einstein considered deflection of light - unaware of Soldner's work - obtained same deflection angle since GR not fully developed

- 1912 Recently it was discovered that Einstein had derived the lens equation, the possibility of a double image and the magnifications of the images

- 1919 a triumph of GR - Eddington - light deflection during eclipse within 20% of prediction

- 1924 Chwolson suggested the possibility of a "fictitious double star" and a ring image in the symmetric case

- 1937 Zwicky: galaxies (extragalactic nebulae) more likely to be lensed than stars. Lens as natural telescope

- 1964 Refsdal showed the time delay between images of the same source constrains the Hubble constant

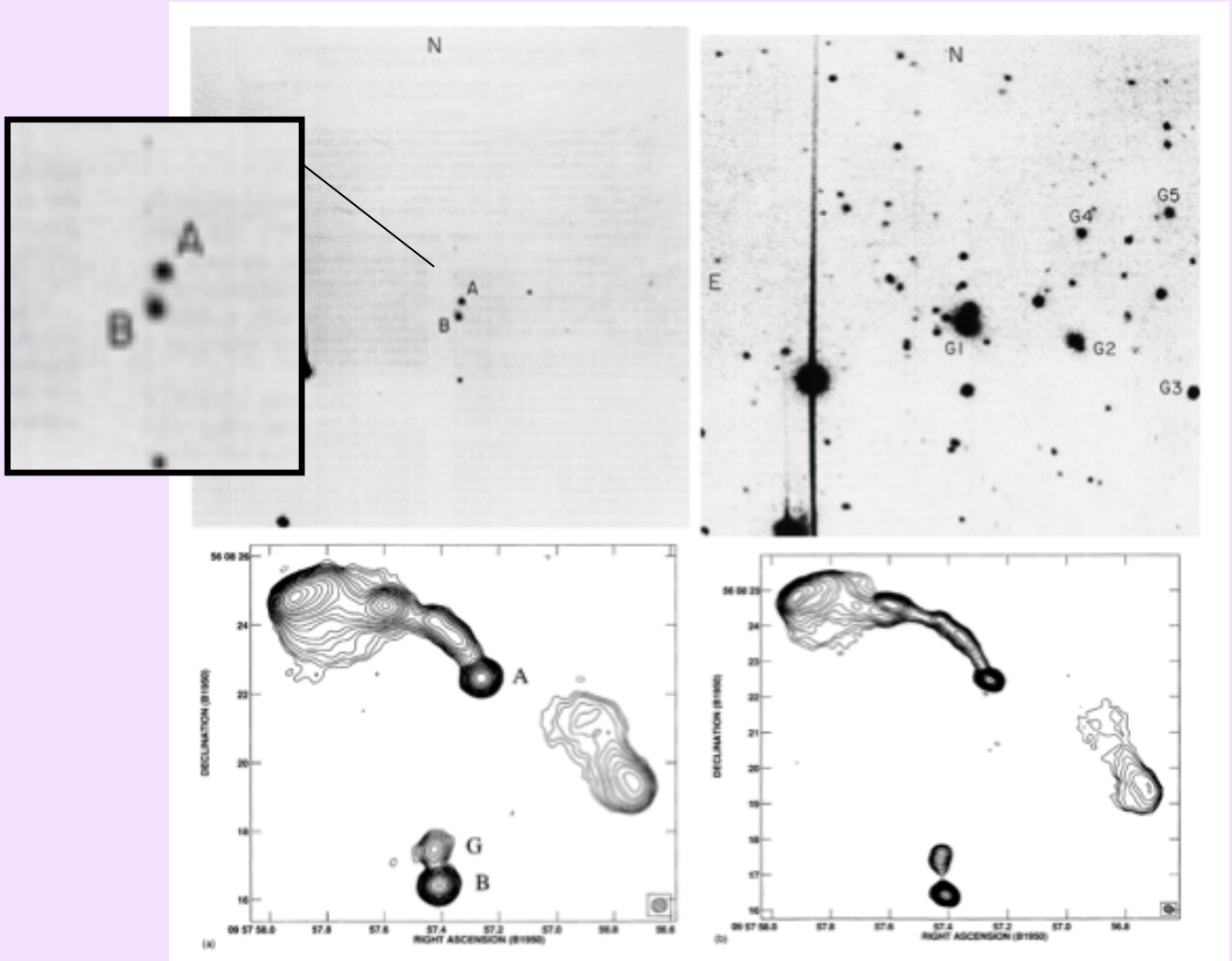
- 1979 first strong lens system discovered - Walsh et al.

- 1990 detection of weak lensing

- 1993 detection of microlensing towards LMC/bulge

- 2000 cosmic shear detected

The first gravitational lens discovered: 0957+561A,B (Walsh, Carswell, Weymann 1979)



Optical

Lensed source is quasar with image sepn. ~6 arcsec

Lens is galaxy aided by cluster

Radio

What can we do with gravitational lensing?

- **Estimate cosmological parameters**

  a tool that complements and cross-checks CMB

- **Constrain mass distributions of galaxies and clusters - test structure formation paradigm**

  how do cluster profiles compare with those from simulations? do dark matter halos exist?

- **Study the relationship between dark & luminous matter -**

  how accurately does light trace mass?

- **Probe the large scale structure of the Universe -**

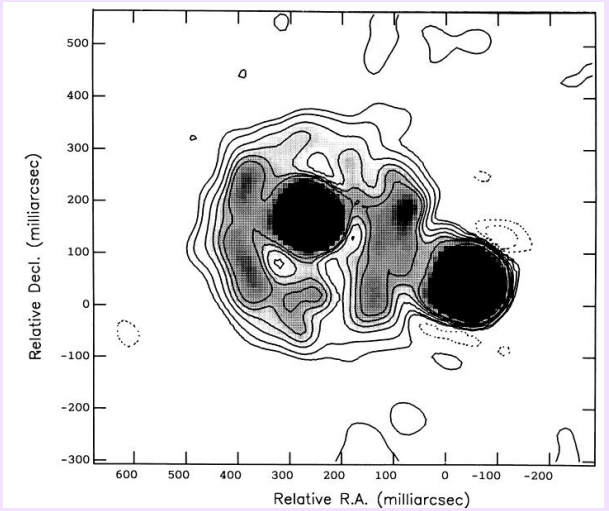
  how do dark matter, and dark energy evolve?

Strong lensing operates on various scales



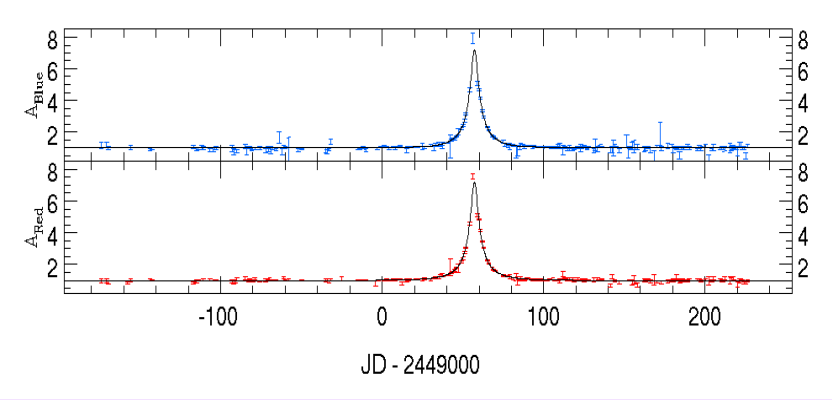
Cluster lens ~30 arcsec

Galaxy lens ~ arcsec

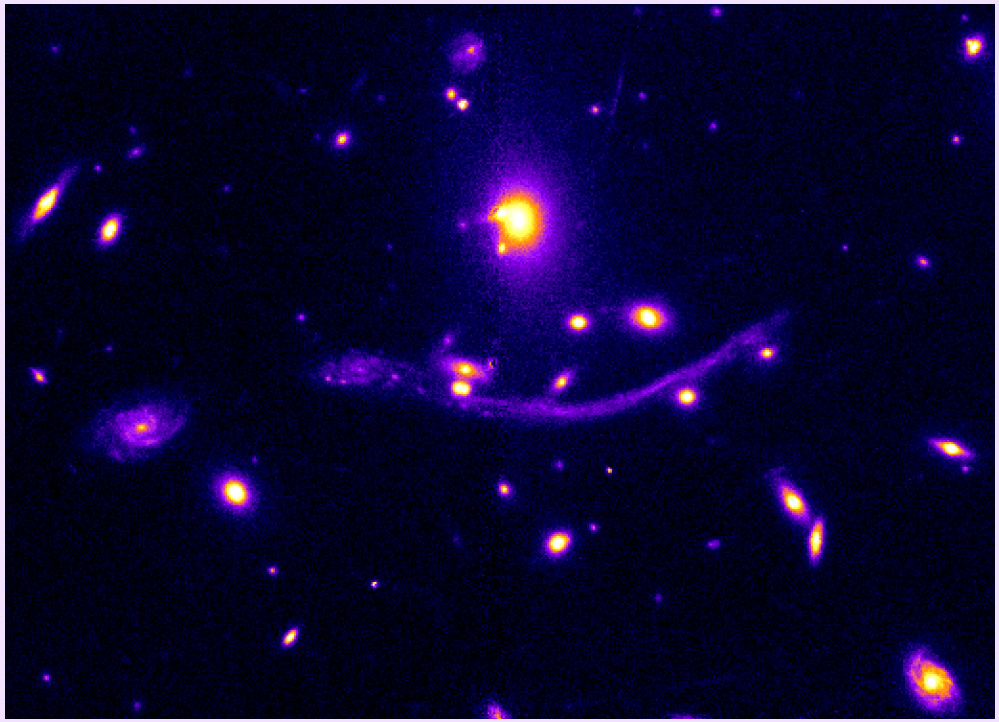


Stellar lens (Galactic microlensing)
~microarcsec

Multiple images unresolved -
transient magnification



MACHO collaboration

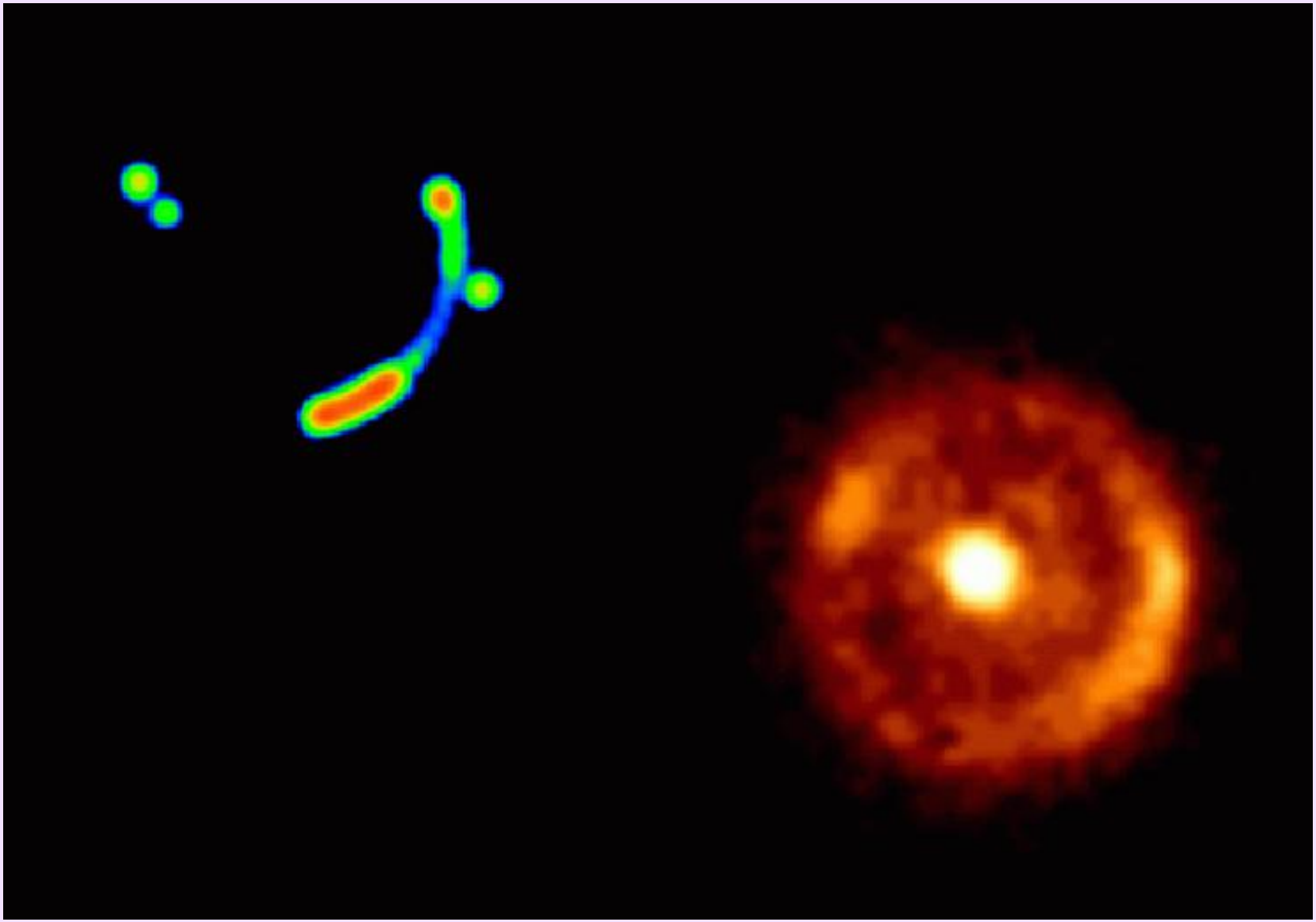


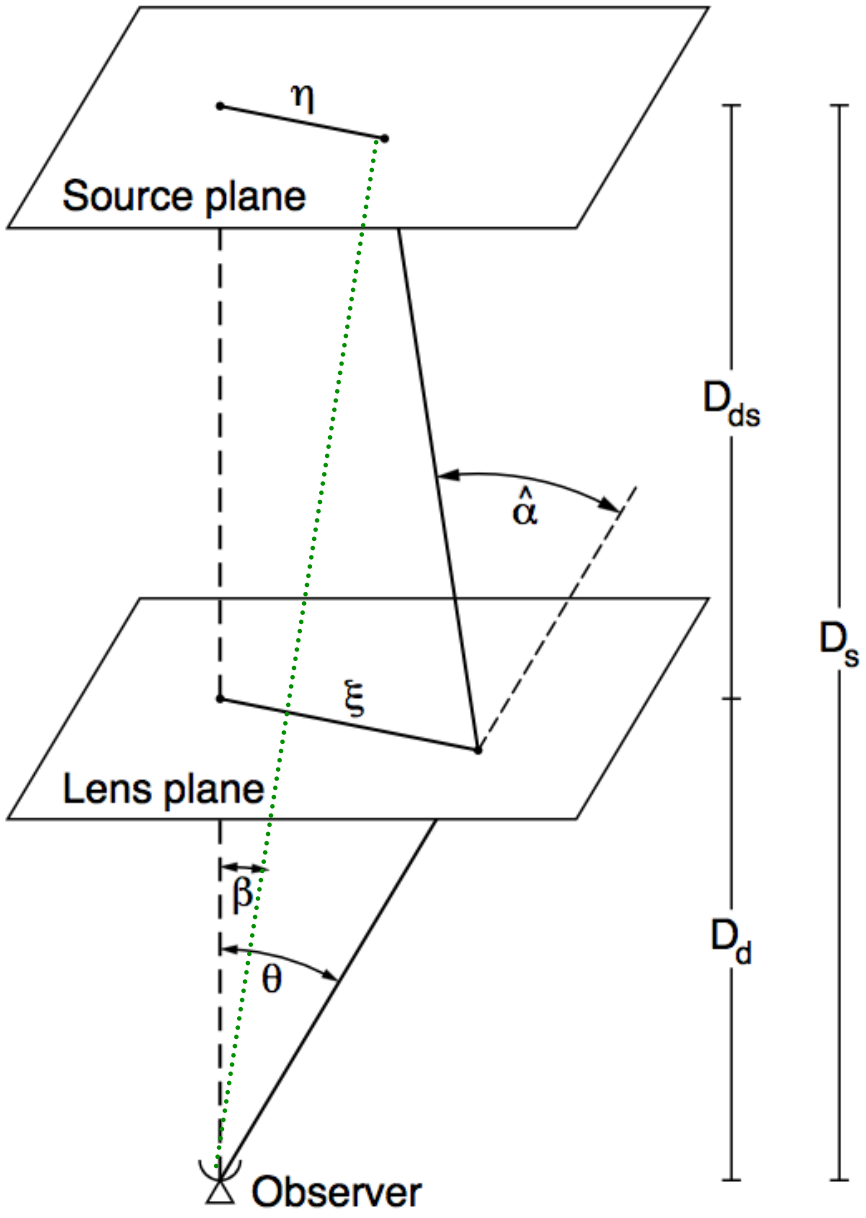
Kneib et al: an arc in Cluster Abell 370; HST

Fruchter: Cluster Abell 2218; HST



King et al: 1938+666 Galaxy scale lens system Radio & HST.





Geometry of lensing

The lens equation

Mapping from $\vec{\beta}$ (source)

to $\vec{\theta}$ (images)

Similar triangles :

$$\frac{\vec{\eta} + \hat{\vec{\alpha}}(\vec{\xi}) D_{ds}}{\vec{\xi}} = \frac{D_s}{D_d}$$

$$\vec{\eta} = D_s \vec{\beta} \quad \text{and} \quad \vec{\xi} = D_d \vec{\theta}$$

$$\rightarrow \vec{\beta} = \vec{\theta} - \frac{D_{ds}}{D_s} \hat{\vec{\alpha}}(D_d \vec{\theta})$$

Defining scaled deflection angle :

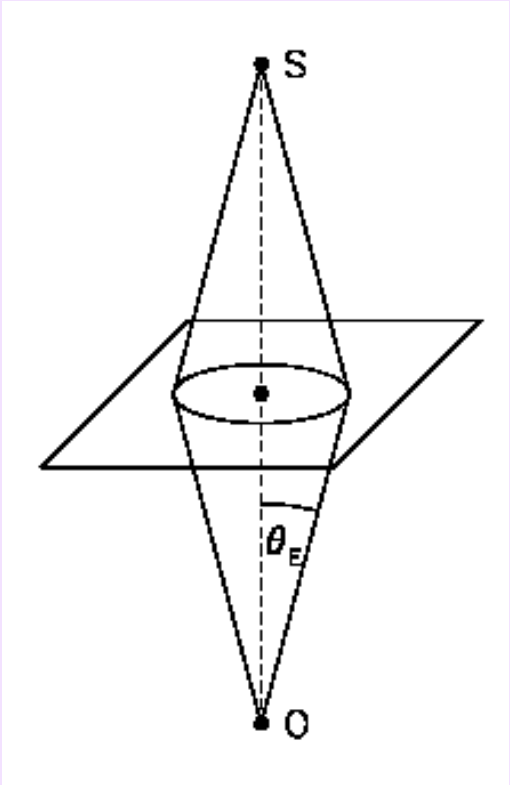
$$\rightarrow \boxed{\vec{\beta} = \vec{\theta} - \vec{\alpha}(\vec{\theta})}$$ NB!

depends on lens mass distribution

Multiple solutions possible!

Lens equation for a point mass lens

$$\hat{\alpha} = \frac{4GM}{c^2 \xi} = \frac{4GM}{c^2 D_d \theta} \Rightarrow \alpha = \frac{4GM}{c^2 D_d \theta} \frac{D_{ds}}{D_s}$$



Lens equation $\beta = \theta - \alpha(\theta)$

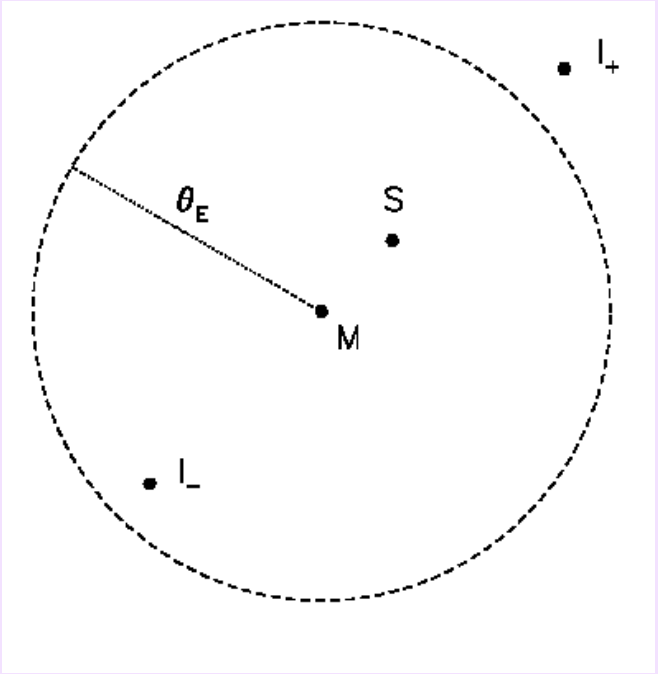
Symmetric case; $\beta = 0$ → Einstein ring

$$\theta_E = \left(\frac{4GM}{c^2 D_d} \frac{D_{ds}}{D_s} \right)^{1/2} = \left(\frac{2R_s D_{ds}}{D_d D_s} \right)^{1/2}$$

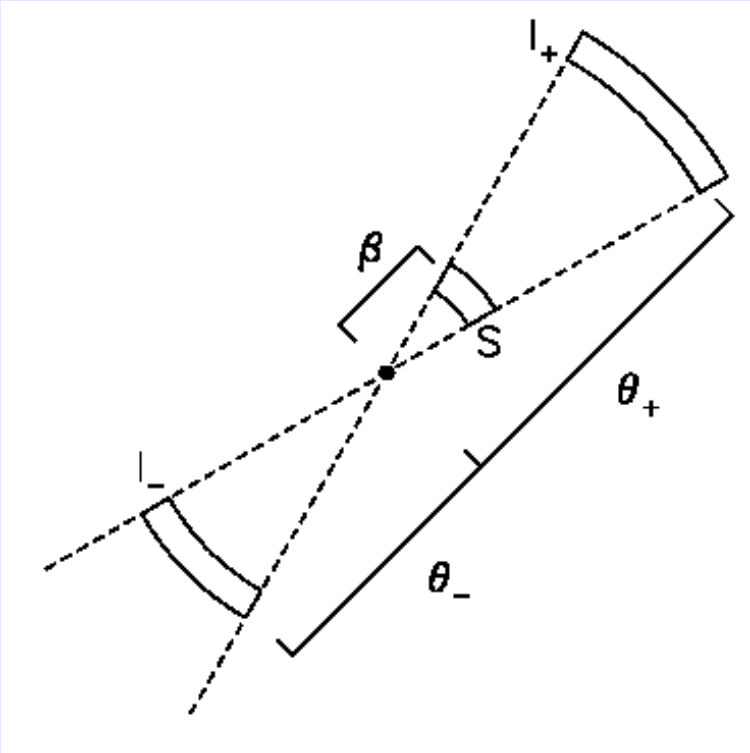
General solution

$$\beta = \theta - \frac{\theta_E^{\,2}}{\theta} \quad \Rightarrow \quad \theta = \frac{\beta \pm (\beta^2 + 4\theta_E^{\,2})^{1/2}}{2}$$

So, the lens equation has 2 solutions -
on either side of the source,
one inside and one outside
the Einstein ring



What are the properties of these images?



magnification= $\frac{\text{image area}}{\text{source area}}$

$$\mu = \frac{\theta}{\beta}\frac{d\theta}{d\beta}$$ subs. for β from previous page

$$\mu = \left[1 - \left(\frac{\theta_E}{\theta}\right)^4\right]^{-1}$$

Magnification of image inside (outside) Einstein ring is negative (positive); negative magnification means an image has opposite parity to the source.

Deflection angle for extended mass distributions

For most situations of astrophysical interest, the extent of the lens << the observer-lens and lens-source distances
- ``**thin lens" approximation**.

Lens mass distribution can be projected onto a “lens plane”, orthogonal to the line-of-sight.

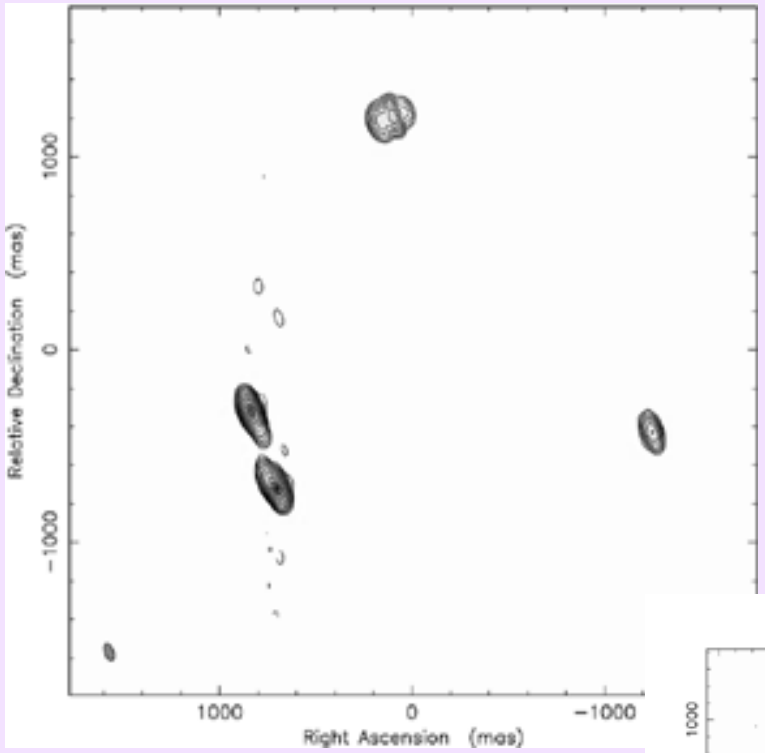
Denoting surface mass density by Σ, the deflection angle is found by integration over the lens plane:

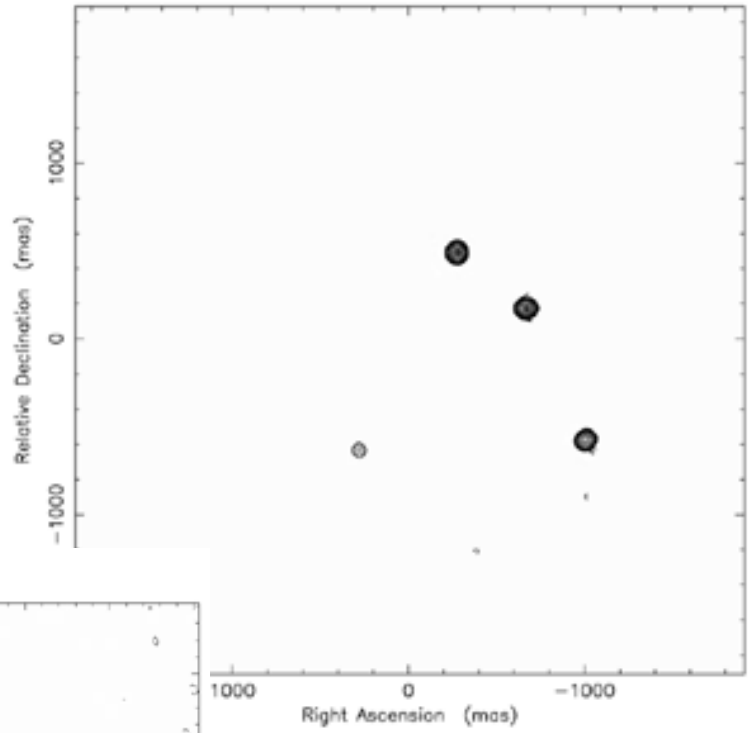
$$\hat{\vec{\alpha}}(\vec{\xi}) = \frac{4G}{c^2}\int_{\mathbb{R}^2} \frac{(\vec{\xi}-\vec{\xi'})\Sigma(\vec{\xi'})}{\left|\vec{\xi}-\vec{\xi'}\right|^2} d^2\xi'$$

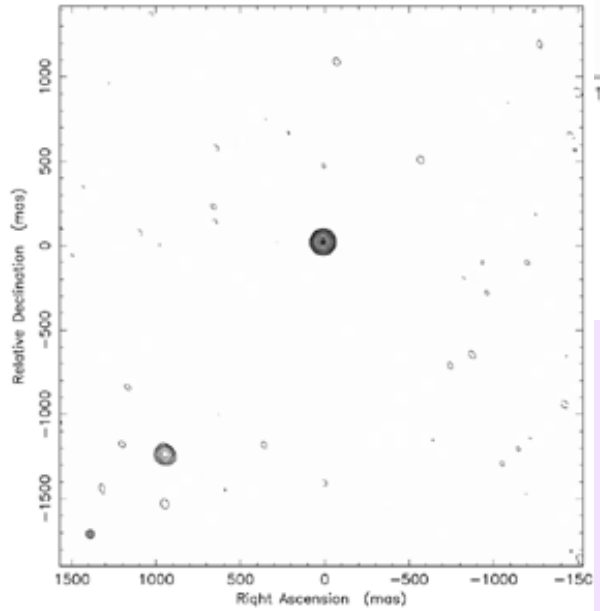
Caustics and critical curves: visualising image formation











Why do we get these multiple image configurations, and how frequent are they?

- Critical curves formally correspond to infinite magnification solutions in the lens plane (det(A)=0).

- In practice, we don't get infinite magnifications!
(i) finite source size;
(ii) break down of geometrical optics.

- The lens equation maps critical curves onto corresponding caustics in the source plane.

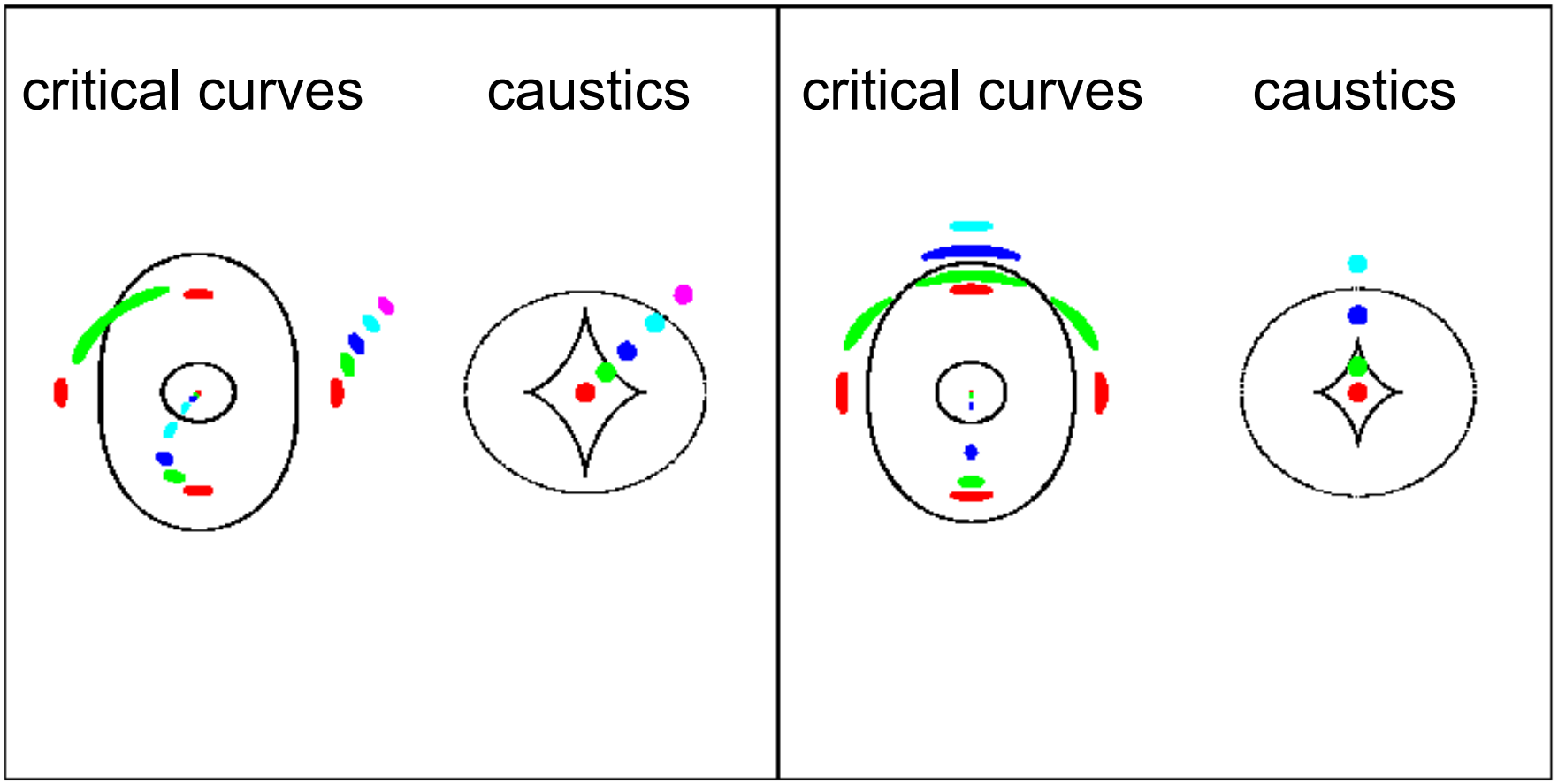
- The location of a source relative to the caustics determines how many images it has, and what they look like.

- A source located just inside a caustic has very magnified images, straddling the critical curve in the image plane.

- As a source crosses a caustic, image pairs are created or destroyed.

- The areas contained within caustics are related to probability for certain number of images.

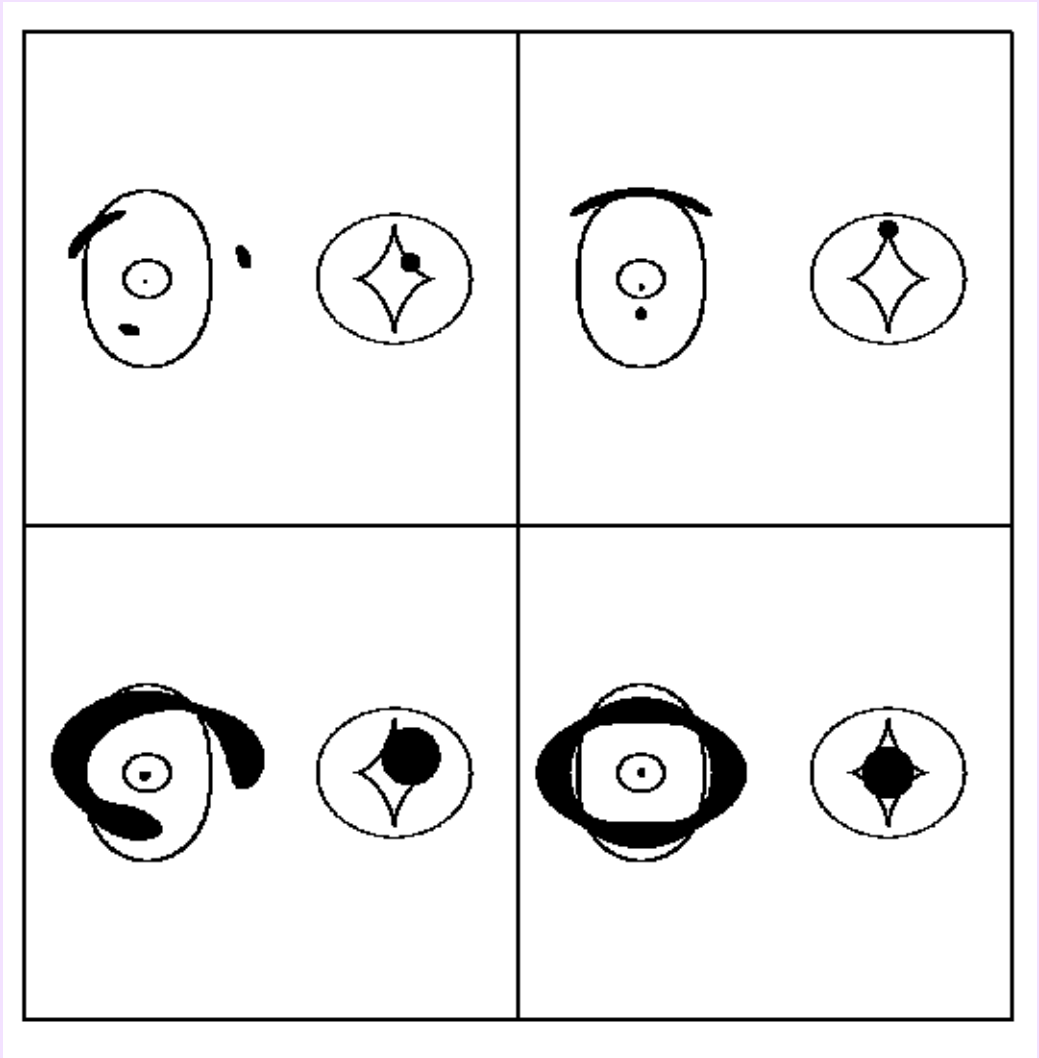
Caustics & critical curves for a typical galaxy lens



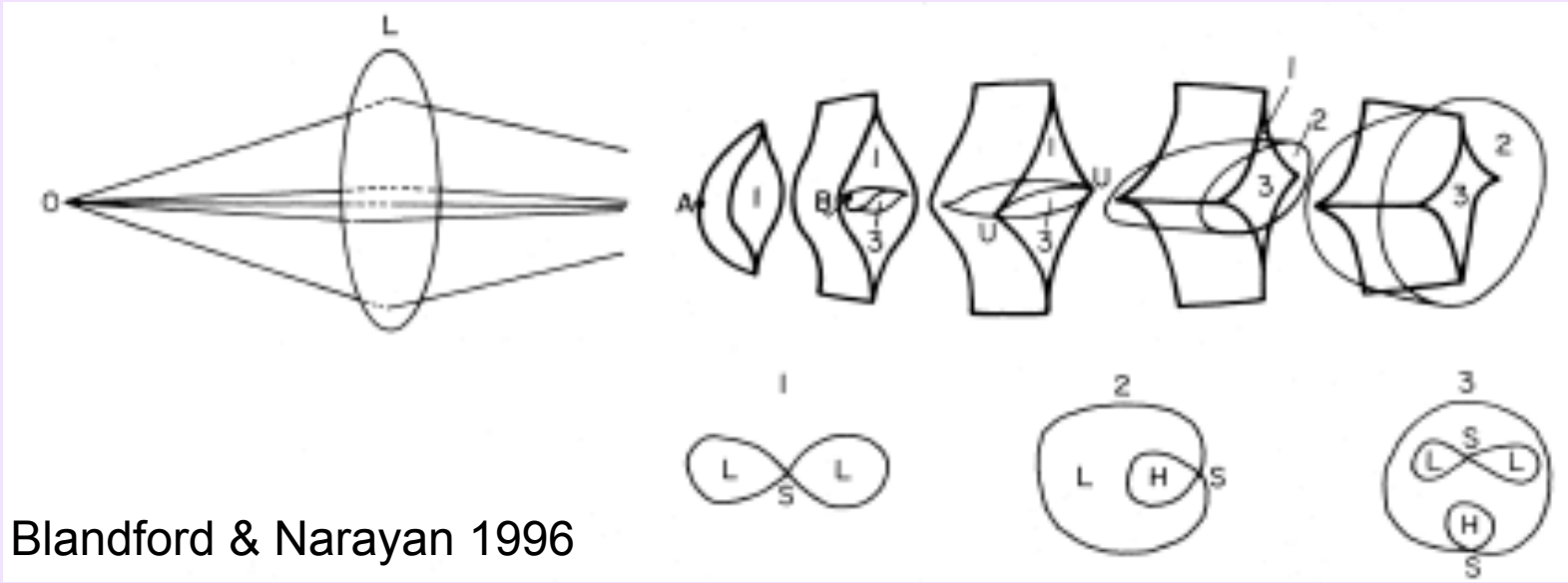
Left panel: source moving towards fold caustic
Right panel: source moving towards cusp caustic

from Narayan & Bartelmann lectures

What happens when the source is large?



Evolution of caustic structure with redshift



Blandford & Narayan 1996

Caustics are actually lines (for cusps) and sheets (for folds) in 3-D; for a fixed observer and lens, location of source determines where we take a “slice” through the caustic structure and thus possible image properties.